



Bucharest University of Economic Studies
Faculty of Cybernetics, Statistics and Economic Informatics
Course: Statistics of Financial Markets



EXAM PROBLEMS – PRACTICE SET

The problems follow the format of the exam: a short computation followed by the interpretation of the result. The data are the daily series used in the course (last day: 18 September 2026). Numerical answers are at the end of the document. Notation: R_t is the simple return, $r_t = \ln(P_t/P_{t-1})$ the log return, and $\text{VaR}_\alpha = -q_\alpha$ (VaR 1% is the loss exceeded on 1% of days).

Useful values: $z_{0.05} = -1.645$, $z_{0.025} = -1.960$, $z_{0.01} = -2.326$, $\varphi(1.960) = 0.0584$, $\varphi(2.326) = 0.0267$, $\ln 2 = 0.693$, $\sqrt{252} \approx 15.87$, $\sqrt{365} \approx 19.10$, $\chi^2_{0.95}(1) = 3.84$, $\chi^2_{0.95}(2) = 5.99$, $\chi^2_{0.95}(10) = 18.31$ (z_p and χ^2_p are quantiles of order p ; φ is the standard Normal density).

Chapter 1. Data, returns and indicators

Problem 1 • The return of a portfolio: Banca Transilvania and OMV Petrom in 2025

The adjusted closing prices were 17.71 lei (31 December 2024) and 25.52 lei (31 December 2025) for Banca Transilvania (TLV), and 0.615 lei and 0.941 lei for OMV Petrom (SNP). On 31 December 2024 an investor bought a portfolio with weights 60% TLV and 40% SNP and held it for the whole year.

- Compute the simple return of each stock in 2025.
- Compute the simple return of the portfolio.
- Compare the log return of the portfolio with the weighted average of the log returns of the two stocks.
- Explain why simple returns aggregate across assets, while log returns aggregate over time.

Problem 2 • The drawdown of the BET index in 2020

Figure 1 shows the BET index and its drawdown between 2015 and 2026. The peak before the COVID-19 crash was 10,219.75 points (23 January 2020), the trough was 7,038.95 points (23 March 2020), and the index regained its peak level on 13 January 2021.

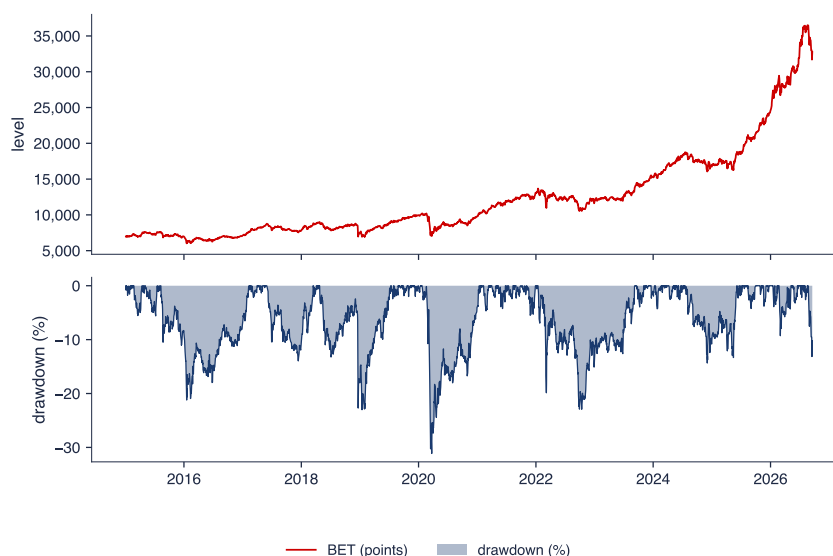


Figure 1: The BET index and its drawdown $D_t = P_t / \max_{s \leq t} P_s - 1$, 2015–2026.

- Compute the maximum drawdown of the episode.
- Compute the percentage gain needed to return from the trough to the peak.
- Interpret the length of the underwater period for an investor with a short horizon.

Problem 3 • The Sharpe ratio and the Sortino ratio: S&P 500

For the S&P 500 (2015–2026, 252 days per year), the annualised mean log return is 11.24%, the annualised volatility is 17.71%, and the annualised downside deviation (threshold $\tau = 0$) is $\sigma_D = 12.82\%$. Take $r_f = 0$.

- (a) Compute the Sharpe ratio.
- (b) Compute the Sortino ratio.
- (c) Explain why the Sortino ratio is larger than the Sharpe ratio for this series.

Chapter 2. Classical distributions and stylised facts

Problem 4 • The Jarque–Bera test for the DAX

For the daily log returns of the DAX index (2015–2026), $n = 2972$, the skewness is $S = -0.565$ and the excess kurtosis is $K = 9.973$.

- (a) Compute the statistic $JB = \frac{n}{6} (S^2 + K^2/4)$.
- (b) Compute the contributions of skewness and of kurtosis to the JB statistic.
- (c) State the decision of the test at the 5% significance level.

Problem 5 • Aggregational Gaussianity: S&P 500

For S&P 500 log returns (2015–2026), the excess kurtosis is 15.85 for daily returns ($n = 2943$), 10.80 for sums over blocks of 5 days ($n = 588$) and 4.00 for sums over blocks of 21 days ($n = 140$). Over the 21-day blocks the skewness is $S = -1.10$.

- (a) Test the normality of the 21-day returns with the Jarque–Bera statistic at the 5% significance level.
- (b) Explain why the excess kurtosis falls as the horizon grows.
- (c) Find the degrees of freedom ν of the Student- t distribution whose excess kurtosis $6/(\nu - 4)$ equals 4.00.

Problem 6 • The worst day of the BET index

On 19 December 2018 the BET index fell by 11.89% (log return). Over 2015–2026 ($n = 2925$ days) the mean daily return is 0.053% and the standard deviation is 0.983%. Over the same period, 42 days had $|z_t| > 3$, where $z_t = (r_t - \bar{r})/s$.

- (a) Compute the standardised return of 19 December 2018.
- (b) Interpret the Normal probability $\Phi(-12.15) \approx 2.8 \cdot 10^{-34}$ of such a day.
- (c) Compare the number of days with $|z_t| > 3$ with the number expected under the Normal distribution ($P(|Z| > 3) = 0.0027$).

Chapter 3. α -stable distributions

Problem 7 • Moments and tails of a stable law

The daily returns of an asset follow an α -stable distribution with $\alpha = 1.5$. For a large threshold x , $P(X > x) \approx C x^{-\alpha}$.

- (a) State whether the mean and the variance of the distribution are finite.
- (b) Compute the ratio $P(X > 2x)/P(X > x)$.
- (c) Compute the ratio $P(X > 10x)/P(X > x)$.

Problem 8 • McCulloch's quantile method: BET

For the daily log returns of the BET index (2015–2026), the empirical quantiles (in %) are $q_{0.05} = -1.383$, $q_{0.25} = -0.380$, $q_{0.50} = 0.085$, $q_{0.75} = 0.543$, $q_{0.95} = 1.417$.

- (a) Compute $\nu_\alpha = (q_{0.95} - q_{0.05})/(q_{0.75} - q_{0.25})$.
- (b) Compute $\nu_\beta = (q_{0.95} + q_{0.05} - 2q_{0.5})/(q_{0.95} - q_{0.05})$.
- (c) Interpret the difference between ν_α and the value 2.44 of the Normal distribution.

Chapter 4. Probability

Problem 9 • Diversification: BET and S&P 500

The standard deviations of daily returns are 0.983% for the BET and 1.116% for the S&P 500 (2015–2026, common days), and their correlation is $\rho = 0.305$.

- (a) Compute the daily standard deviation of the portfolio with weights 50% and 50%.
- (b) Compute the ratio of this standard deviation to the weighted average of the two standard deviations.

- (c) Compute the annualised volatility of the portfolio (252 days).

Problem 10 • The AR(1) process: S&P 500

The first-order autocorrelation of daily S&P 500 returns (2015–2026) is -0.13 . Returns are modelled by the AR(1) process $X_t = \phi X_{t-1} + \varepsilon_t$, with $\phi = -0.13$ and $\text{Var}(\varepsilon_t) = \sigma^2$.

- Compute the second-order autocorrelation.
- Compute the ratio $\text{Var}(X_t)/\sigma^2$.
- Suggest an explanation for a negative first-order autocorrelation of a stock index.

Chapter 5. Heavy tails and extreme value theory

Problem 11 • A Hill plot for Bitcoin

Figure 2 shows the Hill estimator $\hat{\alpha}_k$ of the tail index of Bitcoin daily losses (2015–2026, $n = 4278$) as a function of the number k of losses used. For $k = 100$, $\hat{\alpha} = 2.69$.

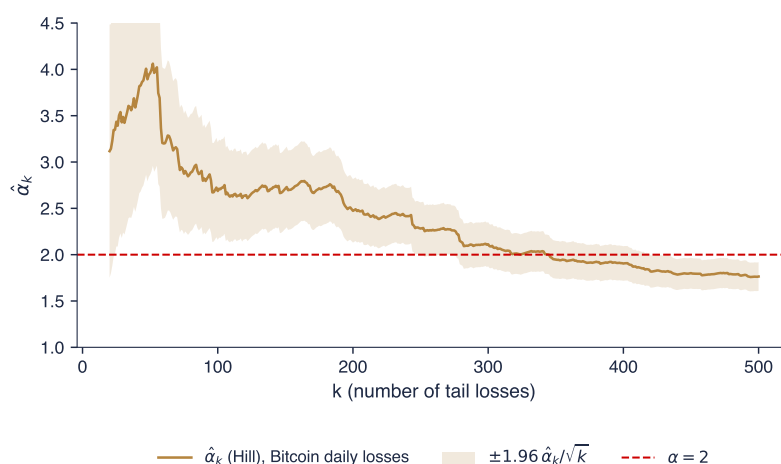


Figure 2: Hill plot: Bitcoin daily losses, 2015–2026.

- Compute the standard error $\hat{\alpha}/\sqrt{k}$ and the 95% confidence interval for $k = 100$.
- State whether, on the basis of this interval, the variance of losses is finite.
- Explain why the estimates for $k = 50$ (3.96) and $k = 400$ (1.90) differ so much.

Problem 12 • A Pareto tail for the S&P 500

For the daily losses of the S&P 500 (2015–2026, in %), the line fitted on the log–log plot of the tail is $\ln \hat{P}(L > x) = -1.391 - 2.753 \ln x$.

- Compute the probability of a daily loss larger than 7%.
- Express the result of (a) as a mean return period, in years of 252 days.
- State whether the variance of losses is finite for this tail index.

Chapter 6. Model selection and risk management

Problem 13 • The Normal or the Student- t distribution for Bitcoin?

On $n = 4278$ daily Bitcoin returns (2015–2026), the maximised log-likelihood is $\ell_N = -11,415.1$ for the Normal distribution (2 parameters) and $\ell_t = -10,745.8$ for the Student- t distribution with location and scale (3 parameters, $\hat{\nu} = 2.19$). Use $\ln 4278 = 8.36$.

- Compute the AIC of the two models.
- Compute the BIC of the two models.
- State what $\hat{\nu} = 2.19$ implies for the existence of kurtosis.

Problem 14 • The Kupiec test

A VaR 1% model records $x = 11$ exceptions in $n = 500$ days.

- Compute the empirical exception rate.
- Compute $LR_{uc} = -2 \ln \frac{(1 - \alpha)^{n-x} \alpha^x}{(1 - \hat{\pi})^{n-x} \hat{\pi}^x}$, with $\hat{\pi} = x/n$.
- State the decision at the 5% significance level.

Chapter 7. Efficient markets, random walk and variance-ratio tests

Problem 15 • Autocorrelations of returns and absolute returns: S&P 500

Figure 3 shows the autocorrelations of daily S&P 500 returns and absolute returns (2015–2026, $n = 2943$). The first five autocorrelations of returns are $-0.128, 0.078, -0.031, -0.062, 0.047$.

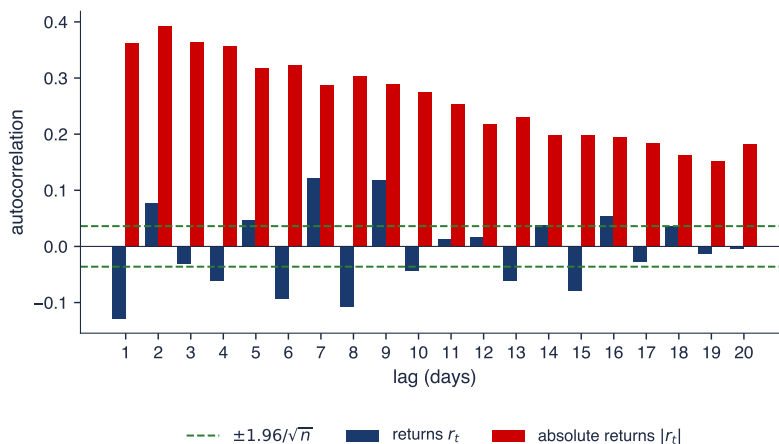


Figure 3: ACF of r_t and $|r_t|$, S&P 500, 2015–2026.

- Compute the limits of the band $\pm 1.96/\sqrt{n}$.
- State how many of the first five return autocorrelations lie outside the band.
- Interpret the difference between the autocorrelations of r_t and those of $|r_t|$.

Problem 16 • The variance ratio from autocorrelations: DAX

For the DAX (2015–2026, $T = 2972$), the first four autocorrelations of daily returns are $-0.020, 0.034, 0.014, -0.014$. Under the random walk with constant variance, the standard deviation of $\widehat{VR}(5)$ is 0.0402.

- Compute $VR(5) = 1 + 2 \sum_{k=1}^4 (1 - k/5) \hat{\rho}(k)$.
- Test the random walk at the 5% level with $z = (VR(5) - 1)/0.0402$.
- Interpret the result in terms of weak-form efficiency.

Chapter 8. Volatility estimators

Problem 17 • Range-based estimators: the S&P 500 on 9 April 2025

On 9 April 2025 (the day a pause in some tariffs was announced), the S&P 500 had $O = 4965.28$, $H = 5481.34$, $L = 4948.43$, $C = 5456.90$, and the previous close was 4982.77. Notation: $u = \ln(H/O)$, $d = \ln(L/O)$, $c = \ln(C/O)$.

- Compute the Parkinson variance $(u - d)^2 / (4 \ln 2)$ and the corresponding daily volatility.
- Compute the Garman–Klass variance $0.5(u - d)^2 - (2 \ln 2 - 1)c^2$ and the corresponding daily volatility.
- Explain why Garman–Klass gives a lower value on a day with a strong trend.

Problem 18 • An EWMA update

Yesterday's EWMA daily volatility was 1.2%, and yesterday's return was -3.5% .

- Compute today's volatility for $\lambda = 0.94$ (RiskMetrics).
- Compute the half-life of the weights, $\ln 0.5 / \ln \lambda$, for $\lambda = 0.94$ and for $\lambda = 0.97$.
- State which of the two values of λ reacts faster to shocks.

Chapter 9. ARCH and GARCH models

Problem 19 • A GARCH(1,1)- t model for the BET

The GARCH(1,1) model with Student- t innovations fitted to daily BET returns (in %, 2015–2026) has $\hat{\omega} = 0.0691$, $\hat{\alpha} = 0.1846$, $\hat{\beta} = 0.7401$.

- Compute the persistence $\alpha + \beta$.
- Compute the long-run variance and the corresponding annualised volatility (252 days).
- Compute the half-life of a volatility shock.

Problem 20 • Multi-step volatility forecasts

With the parameters of the previous problem, the conditional variance for tomorrow is $\sigma_{t+1}^2 = 4$ (volatility of 2%).

- Compute the forecast $E_t[\sigma_{t+10}^2] = \bar{\sigma}^2 + (\alpha + \beta)^9(\sigma_{t+1}^2 - \bar{\sigma}^2)$.
- Compute the 10-day volatility as the square root of the sum of the forecasts $E_t[\sigma_{t+h}^2]$, $h = 1, \dots, 10$.
- Compare the result with the rule $\sqrt{10} \sigma_{t+1}$.

Chapter 10. VaR, ES and backtesting

Problem 21 • VaR and ES for Banca Transilvania

For the daily log returns of Banca Transilvania (2015–2026), the mean is 0.106% and the standard deviation is 1.685%. Historical simulation gives VaR 1% = 4.58% and ES 2.5% = 4.93%.

- Compute VaR 1% and ES 2.5% under the Normal distribution, with $ES_\alpha = -\mu + \sigma\varphi(z_\alpha)/\alpha$.
- Express the historical VaR 1% in lei for a position of 50,000 lei, using $W(1 - e^{-\text{VaR}/100})$.
- Explain the difference between the results of the two methods.

Problem 22 • Clustered exceptions: the Christoffersen test

A VaR 1% model has 10 exceptions in 500 days. Of the 10 exceptions, 3 came right after another exception. The transition counts are $n_{00} = 482$, $n_{01} = 7$, $n_{10} = 7$, $n_{11} = 3$ (n_{ij} : days in state j preceded by a day in state i ; 1 means an exception).

- Compute $\hat{\pi}_{01} = n_{01}/(n_{00} + n_{01})$.
- Compute $\hat{\pi}_{11} = n_{11}/(n_{10} + n_{11})$.
- State the decision of the independence test, given $LR_{ind} = 12.43$.

Chapter 11. Fractal markets hypothesis and long memory

Problem 23 • The Hurst exponent and risk scaling

The average R/S statistic of a return series is 3.2 for blocks of $n = 10$ days and 42 for blocks of $n = 1000$ days.

- Compute the Hurst exponent as the slope in log–log coordinates.
- Compute the memory parameter $d = H - 0.5$ of the corresponding ARFIMA(0, d , 0) model.
- Compute the 20-day VaR 1% from a daily VaR 1% of 2.5%, with the scaling h^H and with the scaling $\sqrt{h}$.

Chapter 12. Scoring models

Problem 24 • Altman's Z-score

A listed manufacturing company has: working capital/total assets $X_1 = 0.12$; retained earnings/total assets $X_2 = 0.25$; EBIT/total assets $X_3 = 0.08$; market value of equity/total liabilities $X_4 = 0.60$; sales/total assets $X_5 = 1.10$.

- Compute $Z = 1.2X_1 + 1.4X_2 + 3.3X_3 + 0.6X_4 + 1.0X_5$.
- Place the company in Altman's zones ($Z < 1.81$ distress zone; $1.81 \leq Z \leq 2.99$ grey zone; $Z > 2.99$ safe zone).
- State the variable that contributes most to the score.

Chapter 13. Machine learning

Problem 25 • Quantile loss for two VaR forecasts

Two models forecast the 5% quantile of the daily return: model A gives $q_A = -2\%$ and model B gives $q_B = -1.5\%$, on each of 4 days. The realised returns were 1.0%, -0.5% , -2.5% , 0.8%. The quantile loss is $\rho_\alpha(u) = u(\alpha - \mathbf{1}\{u < 0\})$, with $u = r_t - q$ and $\alpha = 0.05$.

- (a) Compute the mean quantile loss of model A.
- (b) Compute the mean quantile loss of model B.
- (c) Explain why the quantile loss, and not the squared error, is used to compare VaR forecasts.

Chapter 14. Crypto assets**Problem 26 • Bitcoin at weekends**

For Bitcoin (2015–2026), the standard deviation of daily returns is 2.675% on weekend days and 3.765% on weekdays. A year has 104 weekend days and 261 weekdays.

- (a) Compute the ratio of weekend to weekday volatility.
- (b) Compute the share of weekend days in the annual variance.
- (c) Explain why weekends are calmer on the Bitcoin market.

Chapter 15. Systemic risk**Problem 27 • Joint bad days: JPMorgan Chase and the S&P 500**

Over $n = 2932$ common days (2015–2026), a “bad day” is a day on which the return is below its own 5% quantile. The S&P 500 had 147 bad days, and on 74 of them JPMorgan Chase also had a bad day.

- (a) Compute the number of joint bad days expected if the two series were independent.
- (b) Compute the conditional probability $P(\text{JPM bad day} \mid \text{S\&P 500 bad day})$.
- (c) Interpret the result in terms of systemic risk.

Answers

1. (a) TLV 44.10%, SNP 52.94%. (b) 47.64%. (c) $\ln(1.4764) = 38.96\%$, against a weighted average of 38.92%: log returns do not aggregate exactly across assets. (d) The simple return of a portfolio is the weighted average of the simple returns; log returns add up over time.
2. (a) -31.12% . (b) $+45.19\%$. (c) Almost one year below the peak (January 2020 – January 2021).
3. (a) 0.63. (b) 0.88. (c) The downside deviation is smaller than total volatility: part of the variability comes from rises.
4. (a) $JB \approx 12,475$. (b) Skewness 158; kurtosis 12,317 (almost all of it). (c) $JB \gg 5.99$: normality is rejected.
5. (a) $JB = 121.7 > 5.99$: normality is rejected at the 21-day horizon too. (b) A sum of several returns moves towards the Normal distribution (central limit theorem), but slowly, because of volatility clustering. (c) $\nu = 5.5$.
6. (a) $z = -12.15$. (b) Under the Normal distribution such a day would occur about once in 10^{31} years: the Normal model badly understates the tails. (c) Expected 7.9 days, observed 42: about 5.3 times more.
7. (a) The mean is finite ($1 < \alpha$); the variance is infinite ($2 > \alpha$). (b) $2^{-1.5} = 0.354$. (c) $10^{-1.5} = 0.032$; for the Normal distribution the ratio is practically zero.
8. (a) 3.03. (b) -0.05 . (c) $3.03 > 2.44$: fatter tails than the Normal distribution ($\alpha < 2$); $\nu_\beta \approx 0$ points to an almost symmetric distribution.
9. (a) 0.848%. (b) 0.81: diversification cuts risk by about 19%. (c) $0.848\% \cdot \sqrt{252} = 13.5\%$.
10. (a) $\phi^2 = 0.017$. (b) $1/(1 - \phi^2) = 1.017$. (c) Overreaction to news corrected the next day; for single stocks, also the bounce of the price between bid and ask.
11. (a) $SE = 0.27$; $CI [2.16, 3.22]$. (b) Yes: the interval lies above 2. (c) For small k the estimator has a large variance; for large k observations from the centre of the distribution enter and bias appears.
12. (a) $e^{-1.391} \cdot 7^{-2.753} = 1.17 \cdot 10^{-3}$. (b) About 853 days, i.e. 3.4 years. (c) Yes: $\alpha = 2.753 > 2$ (moments of order below α are finite).
13. (a) 22,834.2 and 21,497.6. (b) 22,846.9 and 21,516.7; both criteria choose the Student- t distribution. (c) $\nu \leq 4$: kurtosis (the fourth moment) is infinite, and the variance exists only just ($\nu > 2$).
14. (a) 2.2%. (b) $LR_{uc} = 5.42$. (c) $5.42 > 3.84$: correct coverage is rejected (too many exceptions).
15. (a) ± 0.036 . (b) Four ($-0.128, 0.078, -0.062, 0.047$). (c) The sign of returns is hard to predict, their size is not: the autocorrelations of $|r_t|$ are large and decay slowly (volatility clustering). The band assumes constant variance, so for r_t it is too narrow.
16. (a) 1.014. (b) $z = 0.36 < 1.96$: the random walk is not rejected. (c) Five-day DAX returns cannot be predicted from past returns: the result is consistent with weak-form efficiency.
17. (a) $3.77 \cdot 10^{-3}$; 6.14%. (b) $1.79 \cdot 10^{-3}$; 4.23%. (c) The term $-(2 \ln 2 - 1)c^2$ lowers the variance when the close is near the high.
18. (a) 1.445%. (b) 11.2 days; 22.8 days. (c) $\lambda = 0.94$.
19. (a) 0.925. (b) $\bar{\sigma}^2 = 0.918$; 0.958% per day; 15.2% per year. (c) 8.9 days.
20. (a) 2.44 (volatility 1.56%). (b) $\sqrt{31.40} = 5.60\%$. (c) $\sqrt{10} \cdot 2\% = 6.32\%$: the rule overstates, because volatility reverts to its mean.
21. (a) $\text{VaR } 1\% = 3.81\%$; $\text{ES } 2.5\% = 3.83\%$. (b) 2236 lei. (c) Historical simulation gives values about 20%–29% higher: the Normal distribution misses the fat tails.
22. (a) $7/489 = 0.014$. (b) $3/10 = 0.30$. (c) $12.43 > 3.84$: independence is rejected; after an exception, a new exception is about 20 times more likely (the model reacts too slowly to volatility).
23. (a) $H = 0.56$. (b) $d = 0.06$. (c) 13.3% with h^H , against 11.2% with $\sqrt{h}$.
24. (a) $Z = 2.22$. (b) Grey zone. (c) X_5 (contribution 1.10).
25. (a) $(0.150 + 0.075 + 0.475 + 0.140)/4 = 0.21$. (b) $(0.125 + 0.050 + 0.950 + 0.115)/4 = 0.31$; model A is better. (c) The quantile minimises the expected quantile loss (VaR is elicitable for this function); the squared error targets the mean, not the quantile.
26. (a) 0.71. (b) $104 \cdot 2.675^2 / (104 \cdot 2.675^2 + 261 \cdot 3.765^2) = 16.7\%$, although weekends are 28.5% of days. (c) Traditional markets are closed: there is no macroeconomic news, no institutional flows and no arbitrage with the ETFs.
27. (a) $2932 \cdot 0.05^2 = 7.3$ days (observed: 74, about 10 times more). (b) $74/147 = 0.50$, against 0.05 without conditioning. (c) The bank suffers together with the market: diversification disappears exactly on bad days, which is what ΔCoVaR and MES measure.